



Practical – 5

Roll No. :	Name:
Class :	Date/Time of Submission: 21-07-2026

Practical No. 5 - Write Algorithm, Code and Output

Q. 1. Write algorithm for the following:

- a. Write a recursive function to print the factorial for a given number.

Code:

```
def fact(n):  
    if n==0 :  
        return 1  
    else:  
        return n* fact(n-1)  
num=int(input("Enter the number : "))  
print("The factorial of a number is " , fact(num) )  
print("Himanshu S037")
```

Output:

```
.PY  
Enter the number : 5  
The factorial of a number is 120  
Himanshu S037
```

- b. Write a function to check the input value is Armstrong or not.

Code:

```
def armstrong(num):  
    sum=0  
    temp=num  
    while (temp>0):  
        digit=temp%10  
        sum+=digit**3  
        temp//=10  
    return sum==num
```

```

temp//=10

if num==sum:
    print(num,"is a armstrong")
else:
    print(num,"is not an armstrong")

num=int(input("Enter the number : "))
armstrong(num)

print("Himanshu S037")

```

Output:

```

Enter the number : 153
153 is a armstrong
Himanshu S037

```

c. Write a function to check the input value is Palindrome or not .

Code:

```

def palindrome(num):
    n= num
    rev=0
    while(num!=0):
        rev=rev*10
        rev=rev+int(num%10)
        num=int(num/10)
    if n==rev:
        print(n,"Is a Palindrome")
    else:
        print(n,"Is not a Palindrome")
num=int(input("Enter the number:"))
palindrome(num)

```

Output:

```

Enter the number:123
123 Is not a Palindrome
Himanshu S037

```

d. Write a lambda function that checks whether a given string starts with a specific character.

Code:

```
starts_with_char=lambda s,c:s.startswith(c)
print(starts_with_char("hello","h"))
print(starts_with_char("hello","t"))
print(starts_with_char("Python","p"))
print("Himanshu S037");
```

Output:

```
True
False
False
Himanshu S037
```

Q. 2. Write a python program to convert temperature from celcius to Fahrenheit. (Use the formula: $\text{fahrenheit} = (\text{celsius} * 9/5) + 32$)

Code:

```
def TemperatureConversion(celsius):
    fahrenheit = (celsius * 9/5) + 32
    return fahrenheit
```

```
celsius=int(input("Enter the value of a temperature :"))
result=TemperatureConversion(celsius)
```

```
print("Celsius to Fahrenheit",result)
print("Himanshu S037")
```

Output:

```
Enter the value of a temperature :32
Celsius to Fahrenheit 89.6
Himanshu S037
```

Q. 3. Write applications of use defined, built-in and external modules in python?